A qualification to an orthogonal rank bound

Human Compression project: F001 source audit

Brooksbank, *On the ranks of string C-group representations for symplectic and orthogonal groups*, arXiv:1912.05512v1, physical page 8, Lemma 4.1, asserts that a sequence of at least six involutions in a subgroup of a five-dimensional orthogonal group satisfying the string commutations must fail the intersection property. Its stated scope includes arbitrary subgroups of the orthogonal group. The following standard permutation-module construction is a counterexample within that scope. It does not disprove the principal theorem concerning $\mathrm{PSp}_4(q)$.

Proposition 1. *There is a nondegenerate quadratic space V of dimension five over $\mathbb{F}_7$ and a subgroup $H \leq O(V)$ having a connected string C-marking of rank six, with every adjacent product of order three. One may moreover take $H \leq SO(V)$.*

Proof. On $M = \mathbb{F}_7^7$ use the coordinate dot product and quadratic form $Q(x) = \sum_{i=1}^7 x_i^2$. Put

$$W = \{x : \textstyle\sum_i x_i = 0\}, \qquad T = \langle (1,1,1,1,1,1,1) \rangle.$$

The dot product on M is nondegenerate, $W = T^\perp$, and $T \subset W$. Hence the radical of its restriction to W is exactly T. Since Q vanishes on T and T is orthogonal to W, the form descends to a nondegenerate quadratic form on $V = W/T$, of dimension five. Coordinate permutations preserve W, T, Q and give an orthogonal representation $\pi : S_7 \longrightarrow O(V)$.

This representation is faithful. If $\pi(g) = 1$, then for every $i \neq j$ the vector $e_{g(i)} - e_{g(j)} - e_i + e_j$ belongs to T. It has at most four nonzero coordinates; a nonzero vector of T has all seven coordinates nonzero. Thus it is zero. Since the characteristic is not two, this forces $g(i) = i$ and $g(j) = j$. Applying this to all pairs gives $g = 1$. The same support argument shows $-1_V \notin \pi(S_7)$: otherwise g would interchange i, j for every distinct pair, which is impossible.

Let $s_i = (i\ i+1)$ for $1 \leq i \leq 6$. The tuple $(\pi(s_1), \ldots, \pi(s_6))$ generates $\pi(S_7)$, consists of involutions and has the required string commutations and adjacent orders three. For completeness, its intersection property can be checked before applying the faithful representation. For any $I \subseteq \{1, \ldots, 6\}$, the subgroup generated by s_i, $i \in I$, is the full product of symmetric groups on the interval components of the path with edge set I. Two such subgroups intersect in the product of symmetric groups on the common refinement of their interval partitions. That refinement consists exactly of the components of the path with edge set $I \cap J$. Therefore

$$\langle s_i : i \in I \rangle \cap \langle s_j : j \in J \rangle = \langle s_k : k \in I \cap J \rangle.$$

Faithfulness transfers this equality to the displayed marking in $O(V)$.

Alternatively use $\widetilde{\pi}(g) = \mathrm{sgn}(g)\pi(g)$. This is again a representation; its kernel is trivial because π is faithful and its image does not contain -1_V. Each $\pi(s_i)$ acts by -1 on the image of $e_i - e_{i+1}$ and fixes its orthogonal hyperplane, so has determinant -1. In dimension five its negative has determinant one. The signed representation is consequently in $SO(V)$ and transfers exactly the same string C-marking. □

Coordinates and the precise failed inference. A basis of V is given by the classes of $e_i - e_7$, $1 \leq i \leq 5$. Its dot-product Gram matrix is $B = I_5 + J_5$, of determinant $6 \in \mathbb{F}_7^\times$. The supplied six literal matrices are computed by permuting these vectors and reducing modulo T. The accompanying exact program checks their group order 5040, all isometry and string identities,

and all 64^2 subset intersection pairs, for both the ordinary and signed representations. These computations corroborate the elementary proof above.

The source proof passes from a nonzero intersection of linear supports to a nontrivial intersection of generated groups. In the ordinary representation, the first and last triples each have a three-dimensional support, and their supports intersect in one dimension. Their groups nevertheless intersect trivially, by the displayed intersection property. Thus that implication cannot justify the stated orthogonal bound.

Scope and attribution. The fully deleted permutation module and adjacent-transposition marking are standard constructions; no priority for either is claimed. This note concerns the exact bound source edition and one auxiliary lemma. It supplies neither a counterexample to the $\mathrm{PSp}_4(q)$ rank spectrum nor a replacement proof of that spectrum. In particular, membership of the signed image in $SO(V)$ does not assert membership in $\Omega(V)$. The latter source theorem, its uniform parameter existence argument and its complete recovery remain pending. There is no open-conjecture credit, compression measurement or Lean claim here.